

Technical Supplement

Listening Without Labels: Open-Set Acoustic Threat Detection for Forest Conservation from Background Audio Alone

Anonymous submission

This supplement provides experimental detail and extended results. The main paper is self-contained; nothing here is required to follow its argument.

1 Dataset and Split Construction

Corpora. Evaluation uses Rainforest Connection rainforest soundscapes (CC BY-NC 4.0), which supply chainsaw and background clips annotated with a recording site. The closed-set training lineage additionally draws on public forest and urban corpora: FSC22 (Bandara et al. 2023), ESC-50/FSD50K-derived negatives, DCASE machine sounds, and Kaggle chainsaw, gunshot, vehicle, and wildfire sets. No model evaluated for a cross-site claim was trained on any RFCx clip; we verified this directly over the training manifests, which contain zero rows with an RFCx source for every public-trained model in Table 2.

Site support. Fourteen RFCx sites appear in the corpus, but most contribute background only. Four carry enough of both classes for a per-site estimate and are used throughout: warsi (412 chainsaw / 198 background test clips), tambopata (41/72), romania (30/15), pooks (10/12). The remaining sites contribute background to training and to the representation probe, which uses seven sites with at least 20 clips each.

Leave-one-site-out construction. For a held-out site S , every clip recorded at S becomes the test split; train and validation are drawn from the remaining sites plus public data. Validation is used only to calibrate the decision threshold, never to select a model.

Leakage audit. RFCx clips are fixed-length windows cut from longer recordings, so filename-level disjointness is insufficient: two windows of one recording could otherwise straddle a split. We therefore key on the source recording, stripping the trailing `<start>-<end>` window index, and verify three properties per fold: zero held-out-site clips in train, zero in validation, and zero shared source recordings between any two splits. All four folds pass on all three counts.

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2 Implementation Details

Detector hyperparameters. Covariance shrinkage $\alpha = 0.1$ toward the diagonal with $\epsilon = 10^{-6}$ ridge; background-CDF quantile grid of 1001 points; similarity floor $\theta = 0.5$; softmax temperature 0.1; false-positive budget $\beta = 0.10$. These are defaults carried unchanged from the deployed platform rather than tuned per fold, which is why the paper reports a single operating point rather than a sweep.

Embeddings. Site-trained CNN: 64-d pre-classifier feature, log-mel input, 16 kHz, 4s clips, 64 mels. AudioSet CNN14: 2048-d, 32 kHz, via `panns_inference` (Kong et al. 2020). BirdNET V2.4: 1024-d, via `birdnetlib` (Kahl et al. 2021) on the LiteRT runtime. All are frozen; no gradient reaches any encoder.

Supervised probe. Logistic regression, `max_iter = 2000`, balanced class weights, on standardized features fit on the training split only. The decision threshold is the $1 - \beta$ quantile of validation-background scores.

Randomness and compute. Bootstrap resampling (1000 draws) and prototype subsampling use fixed seeds. Experiments run on CPU/Apple-Silicon under macOS with PyTorch 2.7; a full regeneration of every number in the paper takes approximately two hours, dominated by embedding extraction, which is cached.

3 Extended Results

Per-site supervised probe. Table 1 expands the main paper’s aggregate probe table into per-site detail with bootstrap intervals. Two observations that the aggregate hides: the site-trained CNN attains high AUROC on the two smallest folds (romania 0.956, pooks 0.917) while collapsing on the largest (warsi 0.604), which is why we weight by n ; and BirdNET’s advantage is concentrated in romania, where it converts a high AUROC into 0.966 event recall at 0.067 false positives, whereas the site-trained CNN at a comparable AUROC converts to only 0.138.

Closed-set lineage across all four sites. The main paper reports the training-recipe comparison on warsi, the highest-support fold. Table 2 gives the same recipes on all four held-out sites at their own validation-calibrated thresholds. The pattern is stable and slightly more nuanced than the single-site view suggests. Hard-negative mining is again the only

Table 1: Per-site supervised leave-one-site-out probe over frozen embeddings, at a 0.10 background-FP budget. Intervals are bootstrap 95% CIs on AUROC. Clip and event recall are measured at the validation-calibrated threshold; “FP” is the realized background false-positive rate on the held-out site.

Embedding	Site	n_{ch}	AUROC [95% CI]	Clip R	Event R	FP
CNN	tambopata	41	0.797 [.71,.88]	0.341	0.341	0.056
CNN	warsi	412	0.604 [.56,.65]	0.396	0.397	0.237
CNN	romania	30	0.956 [.86,1.0]	0.133	0.138	0.000
CNN	pooks	10	0.917 [.76,1.0]	0.100	0.100	0.000
AudioSet	tambopata	41	0.717 [.60,.82]	0.537	0.537	0.167
AudioSet	warsi	412	0.785 [.75,.82]	0.595	0.594	0.167
AudioSet	romania	30	0.900 [.80,.98]	0.733	0.759	0.133
AudioSet	pooks	10	0.625 [.36,.86]	0.400	0.400	0.250
BirdNET	tambopata	41	0.737 [.63,.83]	0.463	0.463	0.056
BirdNET	warsi	412	0.777 [.74,.81]	0.502	0.505	0.167
BirdNET	romania	30	0.973 [.91,1.0]	0.933	0.966	0.067
BirdNET	pooks	10	0.833 [.63,.99]	0.600	0.600	0.167

Table 2: Closed-set training lineage, deployment-calibrated chainsaw recall (R) and background threat-false-positive rate (F) per held-out site. Every model here contains zero RFCx clips in training.

Recipe	tambopata		warsi		romania		pooks	
	R	F	R	F	R	F	R	F
baseline	0.000	0.028	0.005	0.030	0.000	0.133	0.000	0.250
+ augmentation	0.000	0.014	0.002	0.066	0.000	0.000	0.000	0.167
+ hard negatives	0.171	0.014	0.495	0.136	0.033	0.067	0.000	0.000
+ more hard neg.	0.000	0.069	0.012	0.197	0.000	0.133	0.000	0.167
+ expanded set	0.000	1.000	0.000	0.854	0.000	0.800	0.000	1.000
+ DCASE machine	0.000	0.861	0.022	0.556	0.000	0.867	0.000	0.917
+ conservative bg	0.000	0.500	0.000	0.040	0.000	0.867	0.000	0.417

lever that moves recall at all, and on tambopata it does hold the budget (0.171 recall at 0.014 FP), but that recall is far below the closed-set model’s own in-domain performance and it does not hold across sites: the same recipe reaches 0.495 on warsi only by spending 0.136 FP, and yields 0.033 and 0.000 on the two small folds. No recipe delivers usable recall while respecting the budget on every unseen site.

Where BirdNET’s label-free detector fails. Under the unsupervised detector, BirdNET flags nothing at any site (0.000 recall) despite threshold-free AUC as high as 0.821 on romania. The cause is visible in the score distribution: BirdNET’s background embedding cloud is tight enough that the $1 - \beta$ quantile of the training-site background Mahalanobis distance lands above nearly all held-out-site scores, so the transferred threshold admits nothing. This is the same operating-point transfer failure diagnosed for the closed-set classifier, arising here without any supervision, and it is why the paper reports recall at a fixed budget alongside AUC rather than AUC alone.

Honest-unknown holdout. Removing chainsaw’s prototype and re-scoring the held-out site’s chainsaw clips leaves flagging unchanged by construction, and 100% of flagged clips are predicted *unknown* with mean unknown mass 1.0 at every site with held-out chainsaws. With prototypes for other classes present, none of that mass is misattributed to them.

4 Reproduction

The code and data supplement contains the detector, both evaluation harnesses, the plotting scripts, all report files backing every number in the paper, the leave-one-site-out manifests, and the two CNN artifacts. Audio is not redistributed: the evaluation corpus is third-party under a NonCommercial license, so we ship manifests naming every clip instead. Every table and figure regenerates from the committed reports without audio; full re-runs additionally require the public corpus. The mathematics is NumPy-only and covered by 123 unit tests that run without the audio stack.

References

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- Kahl, S.; Wood, C. M.; Eibl, M.; and Klinck, H. 2021. BirdNET: A deep learning solution for avian diversity monitoring. *Ecological Informatics*, 61: 101236.
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